

Computer Network — Lab Sessions

Lab Session Time & Location:

Session	Week	Details	Tasks
1	14	Monday AM D3-b319	Task 1 Demo & QA
2	14	Wednesday AM D3-b319	Task 2 Demo & QA
3	16	Monday AM D3-b319	Task 3 Demo & QA
4	16	Wednesday AM D3-b319	Task 4 Demo & QA

Submissions:

The lab session tasks **SHOULD** be demoed in the sessions.

A report is required to be submitted after the completion of all the tasks.

Your report should follow the listed requirements:

1. Using university lab report template
2. Group work(2-4 members per group), but individual report.
3. NO PLAGIARISM.
4. Store your report in a fold named as “studentID-name” (for example “20240123456789-张三”).

Deadline:

Before the exam date

Task 1 Network Traffic Analysis

Purposes and tasks:

- (1) Learn the use of Wireshark. You can do this task on your own PC and own LAN.
- (2) Learn analysis of network traffic caught.
- (3) Explain TCP hand-shaking process.
- (4) Learn how TCP hand-shaking failure can be handled.
- (5) Analysis ARP packets

Steps:

1. Setting up experimental environment

- (1) Wireshark
- (2) PC connected to network (LAN or WLAN)

2. Basic operations on Wireshark

- (1) Monitor on a particular network interface
- (2) Capture particular packets.

3. Record an example of a successful TCP three-hand-shaking

4. Record an example of hand-shaking failure

5. Monitoring ARP packets

6. Monitoring ICMP packets

Task 2 Basic Router Operations & Static Routing

1. Experimental Purposes

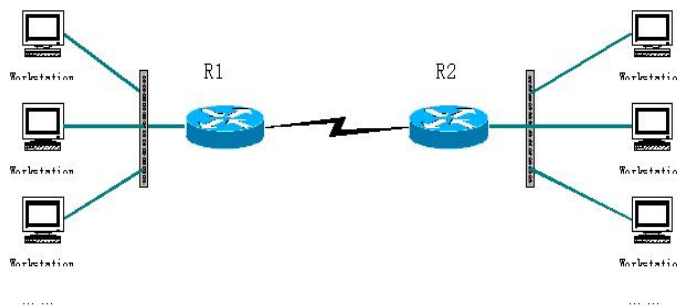
- (1) Get familiar with Cisco PacketTracer and router
- (2) Learn basic router operation command
- (3) Static routing

2. Please answer following questions in your lab report

- (1) What is a router
- (2) How can we remotely log onto a router?
 - What tools can be used?
 - How could you configure these tools?
- (3) What are the different working modes of a router?
 - Prompts
 - Meaning/function
 - How can you switch between different modes?
- (4) Describe the steps/commands of setting IP address on a router interface
- (5) Explain how 'show' command can be applied to
 - obtain router information
 - interface information/configuration
- (6) Describe how routing entries can be manually configured.

2. Experimental Environment

- (1) Network topology (for example)



- (2) Experiment devices

Two routers, two switches (optional), two PCs and cables.

- (3) Experiment requirements

Given 3 C-class Network addresses: 192.168.1.0, 192.168.2.0, 192.168.3.0. (For example only. You can design the networks as you like.)

- A. Using the above topology to design network, explain the address for each node and interface.

Please describe the commands/steps to configure these IP addresses

- B. Using static routing, to enable the workstations to communicate with each other.

Explain the routing entries on each router

Explain the commands/steps to configure routing entries

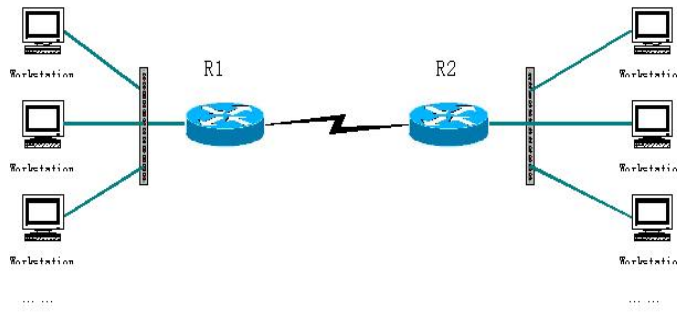
Task 3 Dynamic Routing

1. Experimental Purposes

- (1) RIP configuration
- (2) OSPF configuration

2. Experimental Environment

- (1) Network topology (for example)



- (2) Experiment devices

Two routers, two switches (optional), two PCs and cables.

- (3) Experiment requirements

Given 3 C-class Network addresses: 192.168.1.0, 192.168.2.0, 192.168.3.0. (For example only. You can design the networks as you like.)

C. Using the above topology to design network, explain the address for each node and interface.

D. Using RIP, to enable the workstations to communicate with each other.

Explain RIP

Explain the commands/steps to configure dynamic routing

E. Using OSPF routing, to enable the workstations to communicate with each other.

Explain OSPF

Explain the commands/steps to configure dynamic routing

- (4) Testing and confirmation

How can you test and confirm that the network is working fine?

Task 4 Socket Programming

1. Project Purpose

- (1) Master the basic principles and methods of Socket network programming
- (2) Understand the underlying principle of operation of Socket deeply

2. Project Content

2.1 Requirement

- 1) Basic: Implement a client and a server similar with ftp protocol; complete the basic function of file transfer.

- List the files/directories
- Change working-directory
- Download file/s
- Upload files/s
- Create/remove files/directories
- Etc...

- 2) Live demo

High-layer encapsulation Socket can not be used (like java class library, MFC and so on)

2.2 Reference Materials

Recommended reference books:

- 《UNIX Network Programming》 Richard Stevens
- 《UNIX Environment Advanced Programming》
- 《TCP/IP Networking technology Volume III (Winsock edition)》

3. Project Environment

- (1) Platform: Linux or Windows
- (2) Developing Language: C or C++
- (3) Developing Environment: no limitation, Visual C++, .NET, Vi, etc.

4. Explanation requirements

- Explain the design of your file transfer program.
- Explain the usage of socket.h or winsock.h